# 210Po in the marine environment with emphasis on its behaviour within the biosphere.

The distribution and behaviour of the natural-series alpha-emitter polonium-210 in the marine environment has been under study for many years primarily due to its enhanced bioaccumulation, its strong affinity for binding with certain internal tissues, and its importance as a contributor to the natural radiation dose received by marine biota as well as humans consuming seafoods. Results from studies spanning nearly 5 decades show that (210)Po concentrations in organisms vary widely among the different phylogenic groups as well as between the different tissues of a given species. Such variation results in (210)Po concentration factors ranging from approximately 10(3) to over 10(6) depending upon the organism or tissue considered. (210)Po/(210)Pb ratios in marine species are generally greater than unity and tend to increase up the food chain indicating that (210)Po is preferentially taken up by organisms compared to its progenitor (210)Pb. The effective transfer of (210)Po up the food chain is primarily due to the high degree of assimilation of the radionuclide from ingested food and its subsequent strong retention in the organisms. In some cases this mechanism may lead to an apparent biomagnification of (210)Po at the higher trophic level. Various pelagic species release (210)Po and (210)Pb packaged in organic biodetrital particles that sink and remove these radionuclides from the upper water column, a biogeochemical process which, coupled with scavenging rates of this radionuclide pair, is being examined as a possible proxy for estimating downward organic carbon fluxes in the sea. Data related to preferential bioaccumulation in various organisms, their tissues, resultant radiation doses to these species, and the processes by which (210)Po is transferred and recycled through the food web are discussed. In addition, the main gaps in our present knowledge and proposed areas for future studies on the biogeochemical behaviour of (210)Po and its use as a tracer of oceanographic processes are highlighted in this review. 